

Def A sequence (a_n) is a Cauchy Sequence if $\forall \epsilon > 0, \exists N \in \mathbb{Z}_{>0}$ s.t. whenever $m, n \geq N$, then $|a_m - a_n| < \epsilon$.

Ex. $\pi := 3.14159265 \dots$

$$a_0 = 3$$

$$a_1 = 3.1$$

$$a_2 = 3.14$$

$$a_3 = 3.141$$

$\vdots$

Claim. This is a Cauchy Sequence

Idea. ① $\epsilon = \frac{1}{10}$

if $N = 2$, then if $n, m \geq 2$

then $a_n = 3.14 \ast \ast \ast \dots$

$$a_m = 3.14 \ast \ast \ast \dots ???$$

$$a_m - a_n = 0.000 \dots 0 ??? < 0.1$$

$$\textcircled{2} \epsilon = 0.00001$$

$$\text{Take } N = 6, \quad a_6 = 3.141592$$

$$a_n = 3.141592 \ast \ast \ast$$

For any $m > n \geq 6$, $a_m - a_n < 0.0000001$

$$\textcircled{3} \epsilon = \frac{1}{3^5} > \frac{1}{10^5}$$

$$\text{Take } N = 6$$

Pf. let $a \in \mathbb{Z}_{>0}$ be large enough so that $\varepsilon > \frac{1}{10a}$

Then take $N = a + 1$

□

Thm. A sequence (a_n) is Cauchy if and only if it converges

Important This thm holds in $\mathbb{R}$ but not in $\mathbb{Q}$

Ex. π

A Cauchy sequence in $\mathbb{Q}$ that doesn't converge.

Pf. ($\Leftarrow$). Assume (a_n) [for n large $|a_n - l| < \varepsilon$] converges

Show that it is Cauchy

Rough work Let $\varepsilon > 0$, want N s.t. whenever $m, n \geq N$, $|a_m - a_n| < \varepsilon$

$$\leq \underbrace{|a_m - l|}_{\varepsilon/2} + \underbrace{|l - a_n|}_{\varepsilon/2}$$

Let $\varepsilon > 0$, and $l = \lim_{n \rightarrow \infty} a_n$

Then $\exists N$ s.t. whenever $n \geq N$, $|a_n - l| < \frac{\varepsilon}{2}$

Let $m, n \geq N$

$$\text{Then } |a_m - a_n| \leq |a_m - l| + |l - a_n|$$

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

$$= \varepsilon$$

So (a_n) is Cauchy.

($\Rightarrow$) Assume (a_n) is a Cauchy

Want (a_n) converges

① Want to apply Bolzano-Weierstrass Theorem $\Rightarrow$ WTS (a_n) is bounded

For $\varepsilon=1$, $\exists N$ s.t. whenever $m, n \geq N$, $|a_m - a_n| < 1$

In particular, $|a_m - a_n| < 1 \quad \forall m \geq N$

$$a_N - 1 \leq a_m \leq a_N + 1$$

$$\text{Let } B = \max\{a_1, a_2, \dots, a_N, a_N + 1\}.$$

Then B is an upper bound of (a_n)

$\Rightarrow (a_n)$ is bounded

$\xrightarrow{\text{BW}} (a_n)$ has a convergent subseq $(a_{n_i})_{i \rightarrow \infty} \rightarrow l$

② Show $\lim_{n \rightarrow \infty} a_n = l$

i.e. $\forall \varepsilon > 0 \quad \exists N$ s.t. whenever $n \geq N$, $|a_n - l| < \varepsilon$

Since (a_n) Cauchy, $\exists N_1$ s.t. whenever

$$m, n \geq N_1, |a_m - a_n| < \frac{\varepsilon}{2}$$

• Since $a_{n_i} \xrightarrow{i \rightarrow \infty} l \quad \exists N_2$ s.t.

$$\text{whenever } n_i \geq N_2, |a_{n_i} - l| < \frac{\varepsilon}{2}.$$

$$\text{Let } N = \max\{N_1, N_2\}.$$

Let $m = n_i > N$ for some i , then for any $n \geq N$,

$$|a_n - l| \leq |a_n - a_m| + |a_m - l| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

So any Cauchy seq. (a_n) converges.

$$\leq \underbrace{|a_n - a_m|}_{\substack{\text{If } m, n \text{ large,} \\ \text{this is small} \\ \text{b/c. } (a_n) \text{ is} \\ \text{Cauchy} \\ \frac{\varepsilon}{2}}} + \underbrace{|a_m - l|}_{\substack{\text{Small if} \\ m \text{ is a} \\ \text{part of} \\ \text{conv. subseq.} \\ \frac{\varepsilon}{2}}}$$

